Homework 28: Linear Transformations – A Revisit

BCS A, E, F, G

Problem.

a. ***Composition of Linear Maps.*** If T is a linear transformation such that

$$T\left(\begin{bmatrix}1\\2\end{bmatrix}\right) = 0, \qquad T\left(\begin{bmatrix}2\\3\end{bmatrix}\right) = 1\,.$$

i. What must a and b such that $T: \mathbb{R}^a \to \mathbb{R}^b$?

ii. How many rows and columns the corresponding matrix of transformation A must have?

iii. Find the standard matrix of linear transformation $T: \mathbb{R}^2 \to \mathbb{R}^2$ that first performs a horizontal shear which transforms e_2 into $e_2 + 2e_1$ (leaving e_1 unchanged), then reflects points through the line $y = x$, and then rotates points by $\pi/2$ radians.

iv. A linear transformation $T: \mathbb{R}^2 \to \mathbb{R}^2$ first reflects points through the x_1 $-$axis and then reflects points through the x_2 $-$axis. Show that T can also be described as a linear transformation that rotates points about the origin. What is the angle of that rotation?

b. ***Invertibility of a Linear Transformation.*** Let $T: \mathbb{R}^2 \to \mathbb{R}^3$ defined by

$$T\left(\begin{bmatrix}x_1\\x_2\end{bmatrix}\right) = \begin{bmatrix}-5x_1 + 9x_2\\4x_1 - 7x_2\\x_1 + 3x_2\end{bmatrix}.$$

i. Find the standard matrix of T.

ii. Show that T is one-to-one transformation.

iii. Determine the span of T? Do T spans $\mathbb{R}^3$? Is T invertible?

Submission Date: October 26, 2024 (Saturday $-$ 11:59PM)

Good Luck